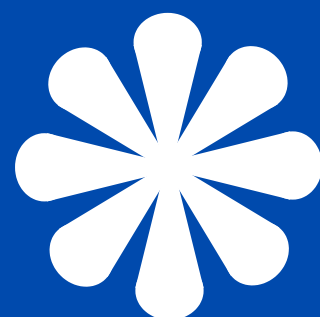




BRAZILIAN ECOMMERCE EDA



OLIST DATASET

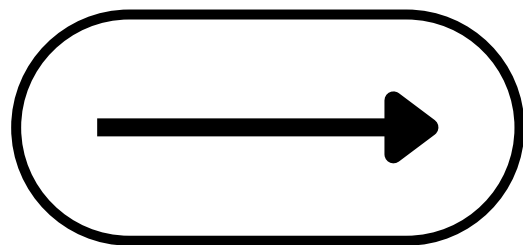
- 2802534084 - Geoffrey Leslie
- 2802544406 - James Tiono
- 2802411664 - Kristoforus Vinan Adriska



DATASET & BACKGROUND

Olist is an e-commerce platform that helps small businesses sell their products on Brazil's largest marketplaces. It acts as a middleman between small retailers and giant websites like Mercado Libre, Amazon, and Magazine Luiza.

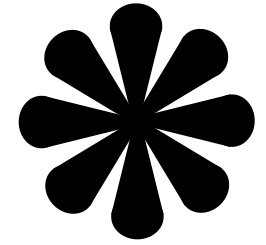
- Connects **small businesses** to major Brazilian marketplaces (like Amazon, Mercado Livre) through one single contract
- Operates like a department store : sellers list products, Olist handles the storefront
- Dataset covers ~99,000 orders across Brazil, 2016-2018



olist



INSIGHT 1: ORDER TRENDS

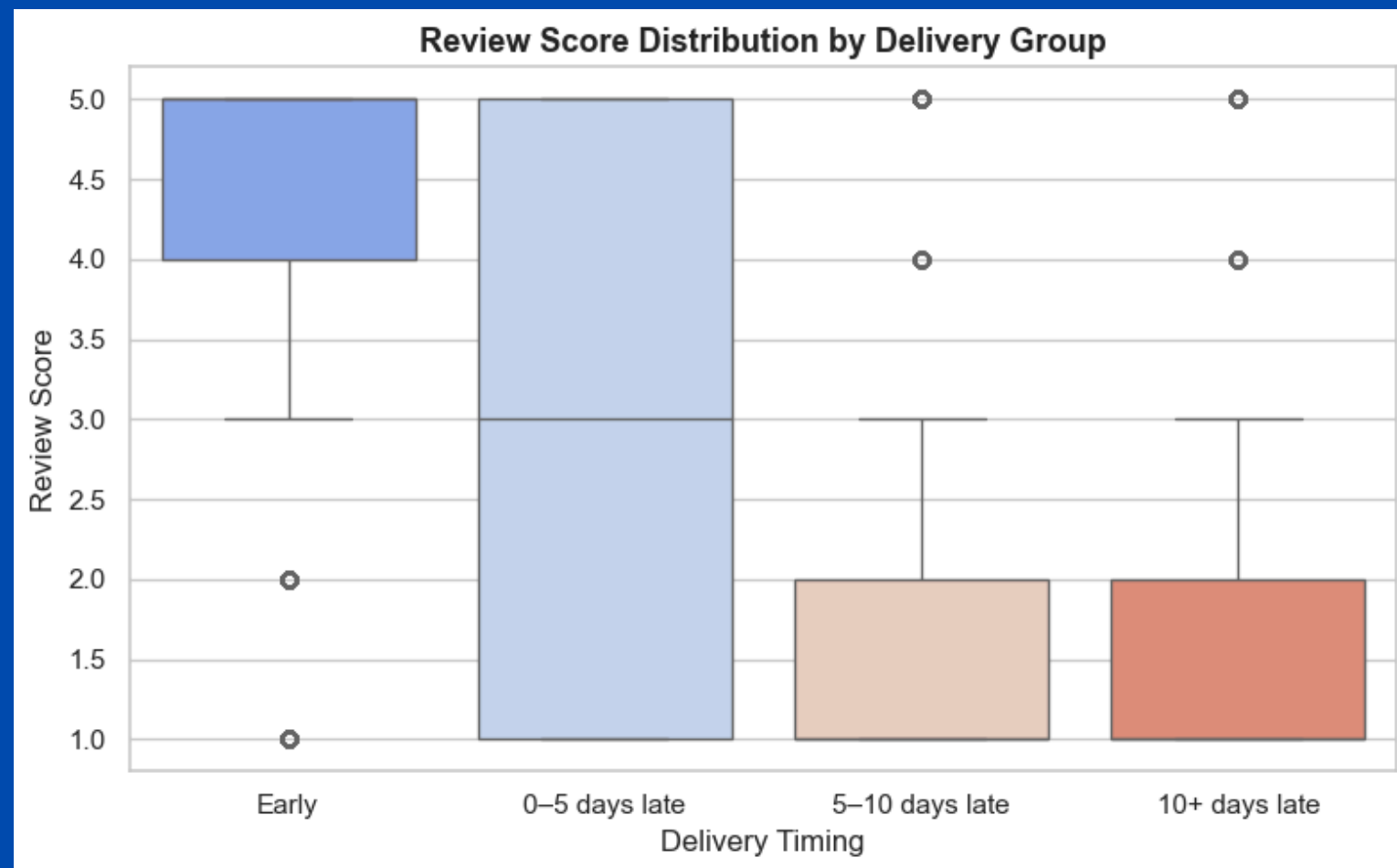


- E-commerce grew steadily from ~300 orders/month in late 2016 to over 7,000/month by November 2017, driven by Black Friday.
- Growth stabilized at ~6,500/month through 2018.

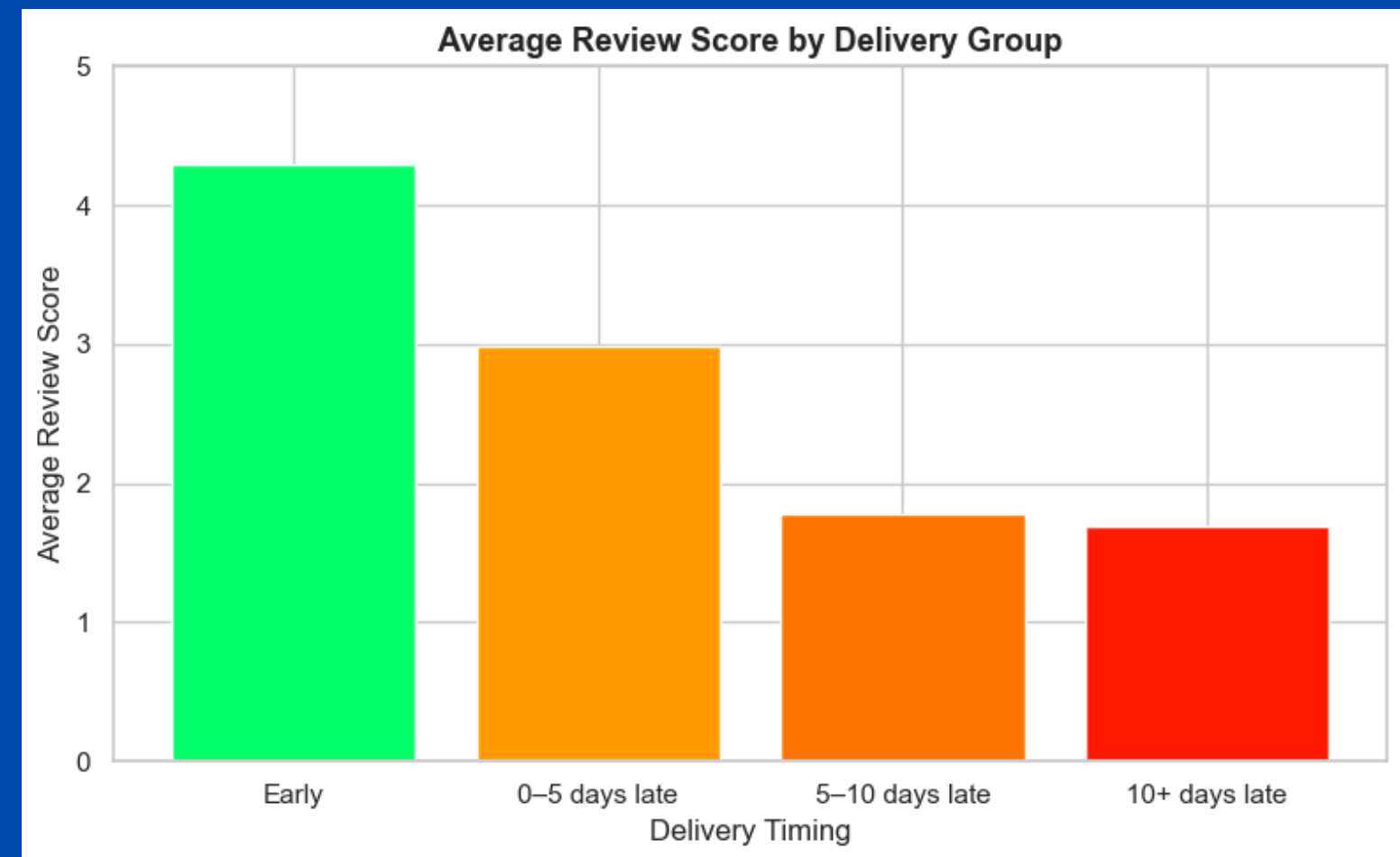


- Brazilians shop most on Mondays and during afternoons (12-18h).
- Weekend orders drop significantly, and barely anyone shops at dawn
- This is useful timing insight for sellers running promotions.

INSIGHT 2: DELIVERY DELAY VS REVIEW SCORE



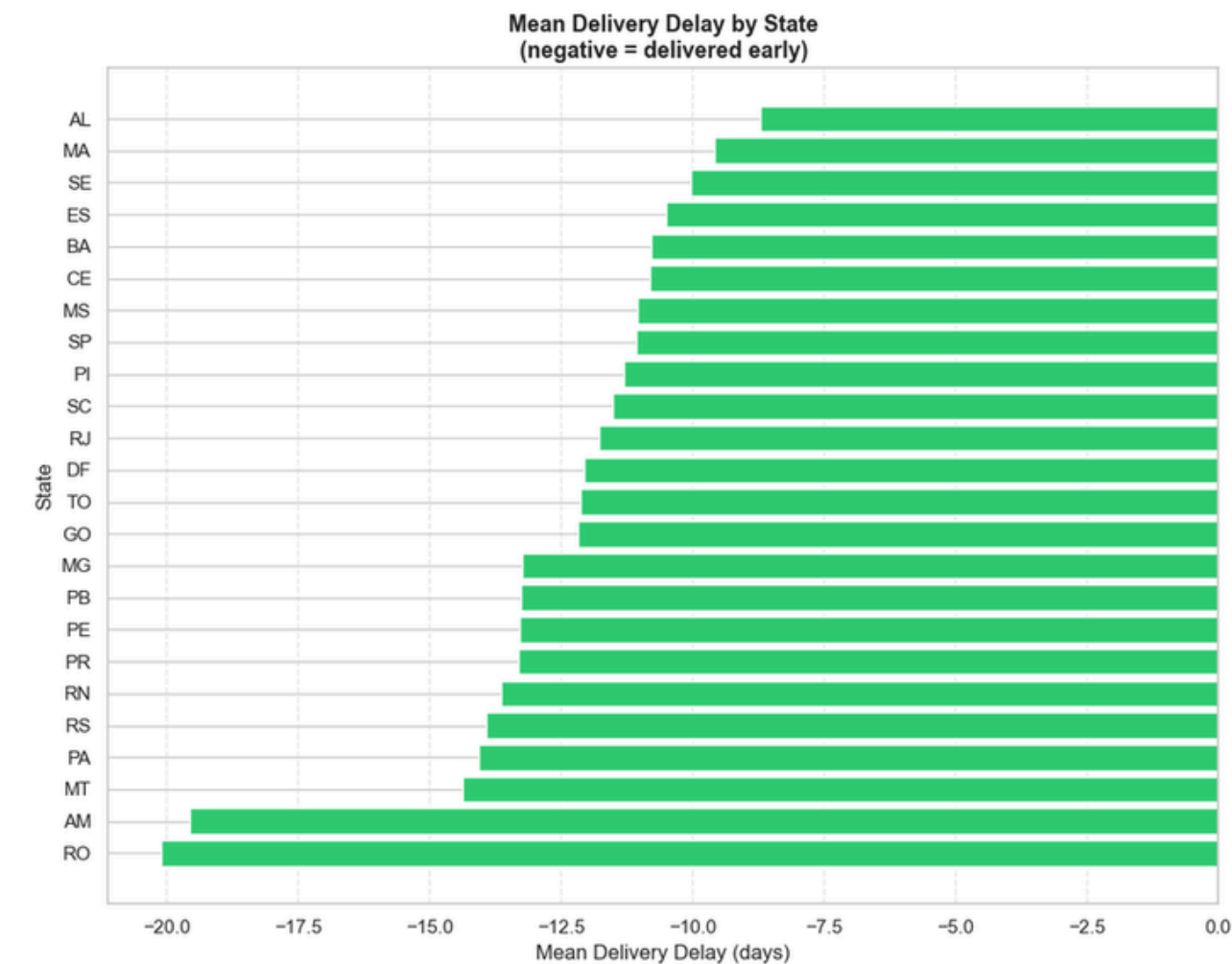
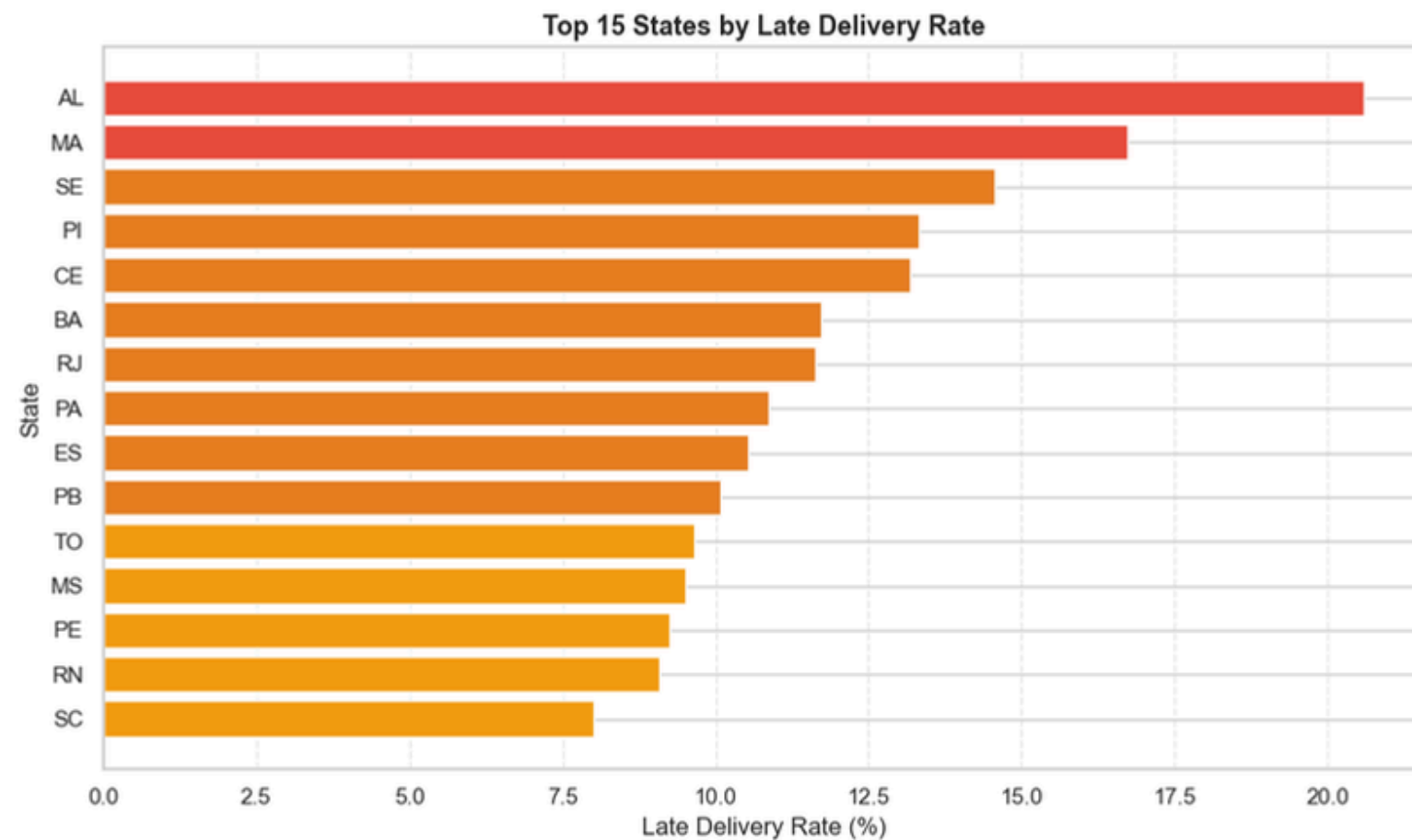
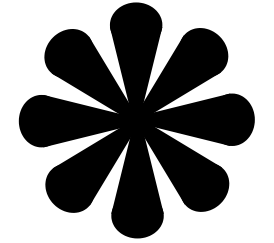
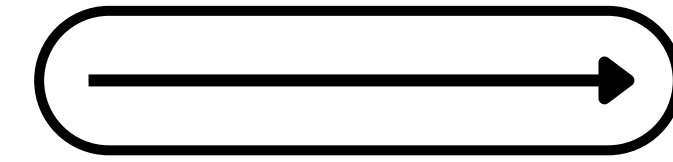
- The spread of scores collapses dramatically as delay increases. Early deliveries cluster around 4-5 stars. Orders 5+ days late are overwhelmingly concentrated at 1-2 stars with almost no high scores.



- Early delivery : 4.3 average score.
- 10+ days late : 1.7 average score.
- A 2.6-point drop purely from delivery timing
- This confirms that logistics efficiency is the single strongest driver of customer satisfaction in this dataset.

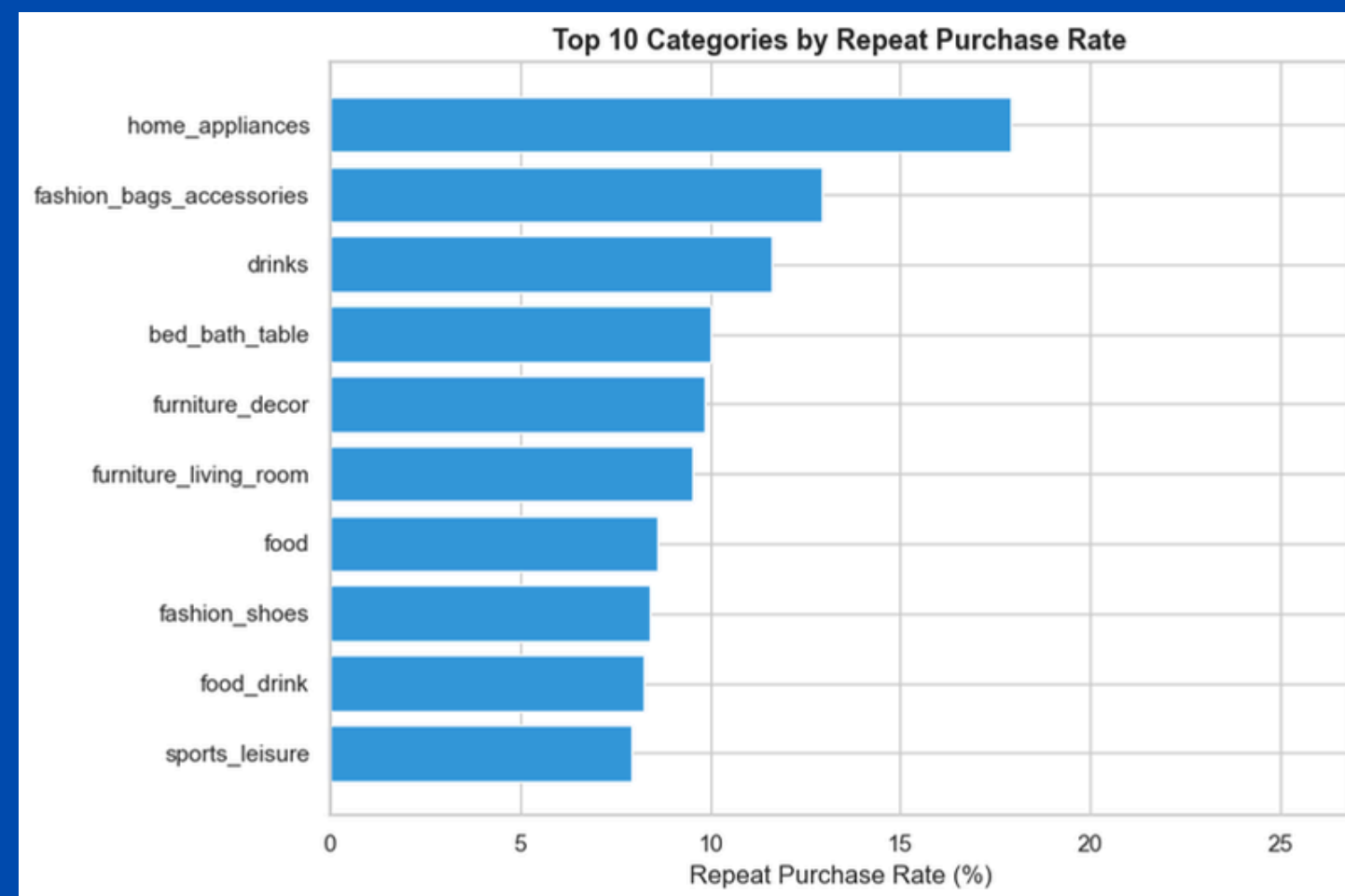


INSIGHT 3: GEOGRAPHIC LATE DELIVERY

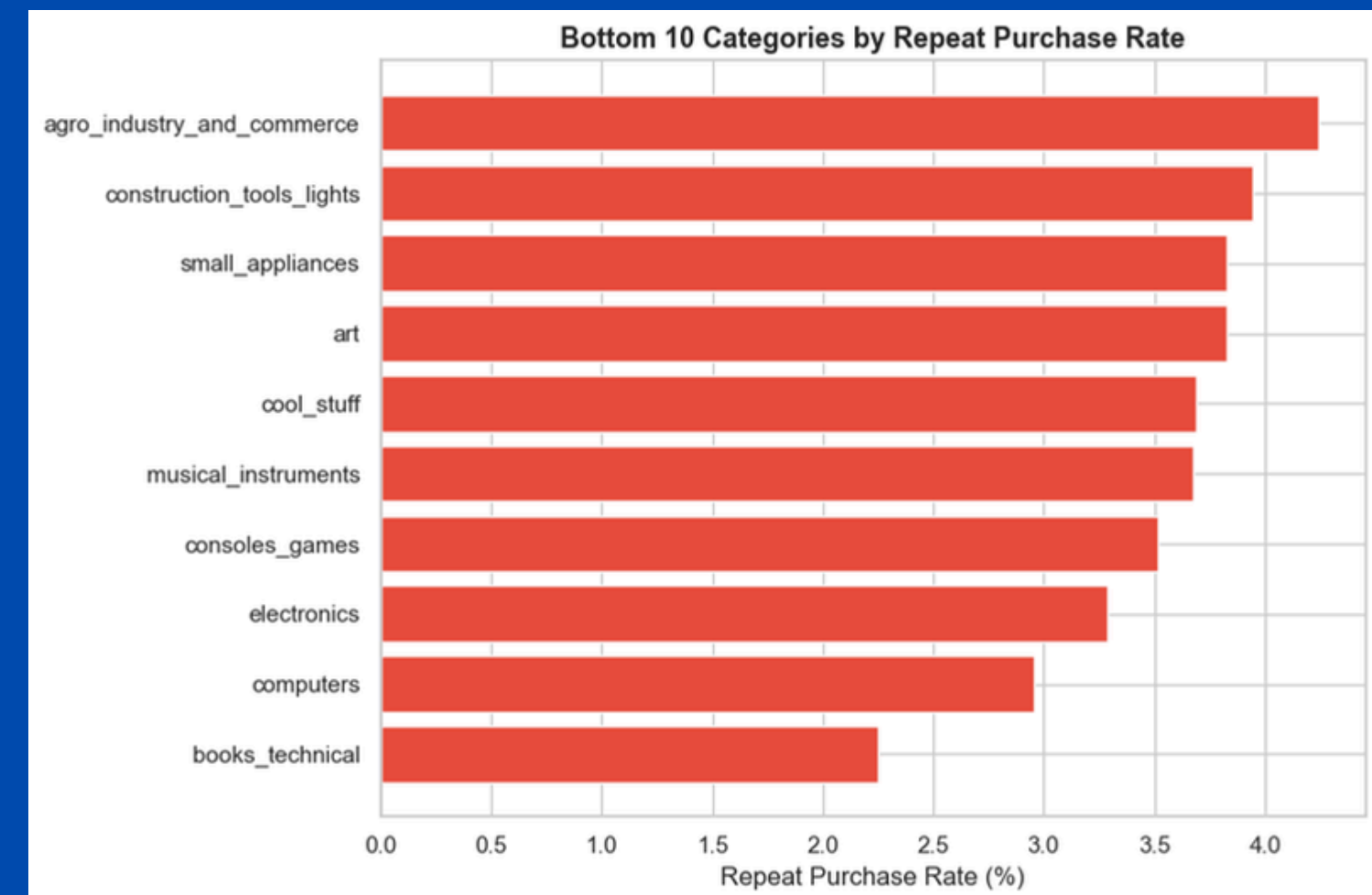


- AL (20.6%) and MA (16.7%) have the worst late delivery rates : both Northeastern states. The Northeast dominates the top of this list, while Southern states like SC barely appear

- All states show negative mean delay
- Meaning on average orders arrive before the estimated date.
- But AL and MA still have the highest late rates because their on-time deliveries aren't early enough to compensate for the ones that are very late.
- RO and AM arrive earliest on average (~20 days ahead) because estimated times are set conservatively for remote regions.



- home_appliances (17.8%) and fashion_bags_accessories (13%) lead repeat purchases
- surprisingly, not food/drinks. This suggests customers who buy lifestyle and home goods are building brand trust with specific sellers on the platform.



- books_technical (2.3%) and computers (3%) sit at the bottom
- these are one-time or infrequent purchases by nature.
- Electronics categories generally show low repeat rates, meaning sellers in these categories rely heavily on acquiring new customers rather than retaining existing ones.

INSIGHT 4: REPEAT PURCHASE



- Overall, only ~3% of all customers placed more than one order meaning the platform is still in acquisition mode, not retention mode.



CONCLUSION

SDG 8:DECENT WORK AND ECONOMIC GROWTH

Finding	SDG 8 Target
E-commerce tripling in 2 years	8.1 : Sustained economic growth
Late deliveries destroy review scores → hurt seller income	8.2 : Higher economic productivity
Northeast states have worst delivery rates	8.1 : Sustained per capita economic growth in developing regions
3% repeat purchase rate : low retention	8.3 : Promote entrepreneurship and small business growth

Brazil's e-commerce market shows strong growth momentum aligned with SDG 8, but logistics inequality and low customer retention remain structural barriers to inclusive economic growth.

Improving last-mile delivery in underserved regions, particularly the Northeast, and building conditions for repeat purchasing are the two highest-impact interventions for making digital commerce a genuine engine of decent work for Brazilian sellers.

